Detecting Fake News

A DS4002 Case Study by Lauren Medica

Every day, billions of people scroll through headlines. Most have never even read the full article. You have probably done it yourself, seen a shocking headline, formed an opinion, and kept scrolling. In an era defined by rapid information sharing and social media, fake news spreads faster than corrections, and often a headline is all it takes to shape someone's view of the world.

Misinformation has real consequences: it influences elections, defines trust in institutions, and in some cases has led to real-world violence. As a data science student, you already have the tools to start tackling this problem. Machine learning can be used to automatically classify news articles as fake or real based purely on their text, no human fact-checker required. But here is the interesting question: does a model need the full article, or is the headline alone enough?

Imagine you are working with a team building a content moderation tool for a news company. Your task is to figure out whether their system needs to process full articles or whether scanning headlines alone is sufficient. Using a labeled dataset of over 40,000 real and fake news articles, you will train two Random Forest classifiers, one on full article text and one on titles only, and compare how well each one performs. Along the way, you will get hands-on practice with text preprocessing, TF-IDF vectorization, and model evaluation using F-1 score. This is the kind of project that shows up in data science internships and jobs, real data, a meaningful question, and results that actually matter. By the end, you will have two trained models and a clear answer to a question that researchers and tech companies are still actively wrestling with.